

BLOOD GROUP SYSTEM

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OBJECT:

- **To find out your own blood group.**

Importance of knowing blood group system

Before transfusion to a patient with massive blood loss

To prevent hemolytic disease of new born (Rh compatibility in newborn)

To solve the legal disputes related to parenting claimant.

To study the Mendelian laws of genetics.

(when mating occurs between two organisms of different traits, each offspring exhibits the trait of one parent only)

When Blood is transfused?

INDICATIONS	CONTRAINDICATION
□ Acute blood loss with hypovolaemia, after RTA. □ During surgery as in cardiovascular bypass surgery □ Exchange of blood in severe anemia and severe hyperbilirubinemia at birth □ Severe anemia, Hb less than 5gm	Risk of overload in patients with □ Chronic anemia □ Cardiac failure

Clinical Significance

- When blood transfusion is necessary, the blood typing of the recipient and the donor should be done very carefully, to avoid hazards of blood transfusion.
- The mismatched blood transfusion will result in **hemolytic jaundice** and transfusion reaction associated with **hemolysis**, **haemoglobinurea** and severe type of **shock**.

History

Austrian immunologist **Karl Landsteiner** discovered the **ABO blood group** System in 1901. In 1910 he won Nobel prize for medicine for this discovery.

In 1940- Karl Landsteiner and Alexander S Wiener reported another **Rh blood group**.

Landsteiner Rule

If an antigen (Ag) is present on a patient's red blood cells the corresponding antibody (Ab) will NOT be present in the patient's plasma, under 'normal conditions'.

What is the principle of blood grouping test?

Principle: The ABO and Rh blood grouping system is **based on agglutination reaction**. When red blood cells carrying one or both the antigens are exposed to the corresponding antibodies they interact with each other to form visible agglutination or clumping.

What is the basis of grouping blood group?

- What makes our blood different from someone else's is the unique combination of protein molecules, called **antigens** and **antibodies**.
- **Antigens** are located on the surface of the red blood cells.
- **Antibodies** are located in plasma.
- **The combination of antigens and antibodies** in your blood is the basis of your blood type.

What are the 4 types of blood groups?

- There are 4 main blood groups (types of blood) – **A, B, AB and O**. Your blood group is determined by the genes you inherit from your parents.
 - Each group can be either RhD positive or RhD negative, which means in total there are 8 blood groups.
- There are **eight** possible types:

A positive	A negative
B Positive	B negative
O positive	O negative
AB positive	AB negative

Basics of ABO Blood grouping

- Based on the presence or absence of antigen A and antigen B, blood is divided into four groups:
- A, B, AB and O
- Blood having antigen A belongs to 'A' group. This blood has β -antibody in the serum
- Blood with antigen B and α -antibody belongs to 'B' group.

If both the antigens are present, blood group is called 'AB' group and serum of this group does not contain any antibody.

If both antigens are absent, the blood group is called 'O' group and both α and β antibodies are present in the serum

Difference between ABO blood group and RH blood group

The main difference between **ABO blood group** and **RH blood group** is that **ABO blood group** is determined by the presence or absence of **A and B antigens** on the surface of the red blood cell whereas **RH blood group** is determined by

the presence or absence of **D antigen** on the surface of the red blood cell

Principle of Blood Grouping

Blood grouping is done on the basis of Agglutination means the collection of separate particles like RBCs into clumps or masses.

- Agglutination occurs if an antigen is mixed with its corresponding antibody which is called **isoagglutinin**, i.e. it occurs when A antigen is mixed with anti-A or when B antigen is mixed with anti-B

Why is O -ve the universal donor?

Type **O-negative** blood does not have any antigens. It is called the "**universal donor**" type because it is compatible with any blood type.

What Is the Universal Recipient Blood Type?

- The universal recipient blood type is **AB positive** (also written as AB+). This means that an AB+ person can safely receive a **blood transfusion with** any of the other blood types.
- Matching the correct blood type is key to a safe blood transfusion or organ transplant.

IMPORTANCE OF ABO GROUPS

- **IN BLOOD TRANSFUSION**

- During blood transfusion, only compatible blood must be used.
- The one who gives blood is called the “donor” and the one who receives the blood is called recipient.
- While transfusing the blood, antigen of the donor and the antibody of the recipient are considered

In mismatched transfusion

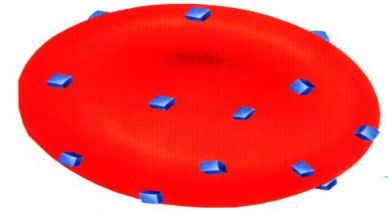
In mismatched transfusion, the transfusion reactions occur between donor's RBC and recipient's plasma.

So, if the donor's plasma contains agglutinins against recipient's RBC, agglutination does not occur because these antibodies are diluted in the recipient's blood.

On the basis of Antigen A or Antigen B , population may be divided in to four blood groups

- A
- B
- O
- AB

💧 A antigen- group A



💧 B antigen- group B



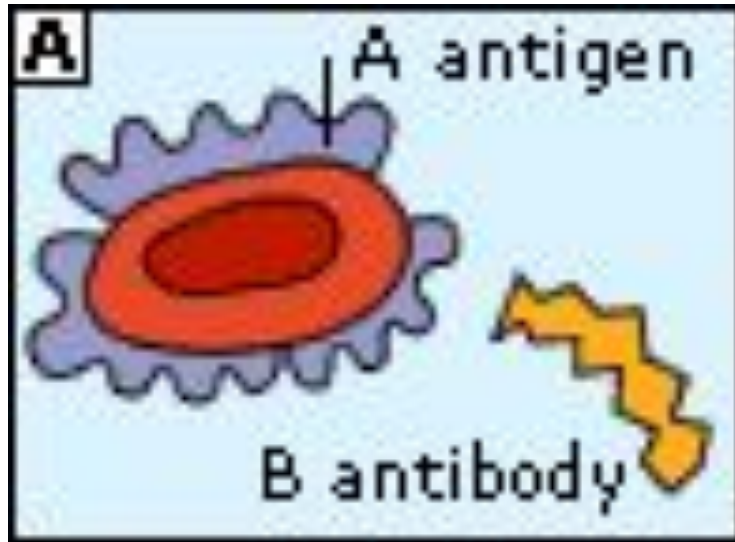
💧 A and B antigen-
group AB



💧 None – group O

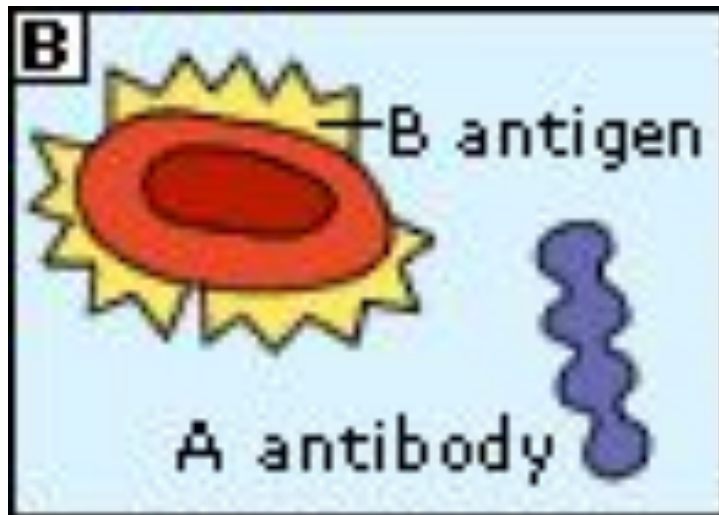


ABO blood grouping system



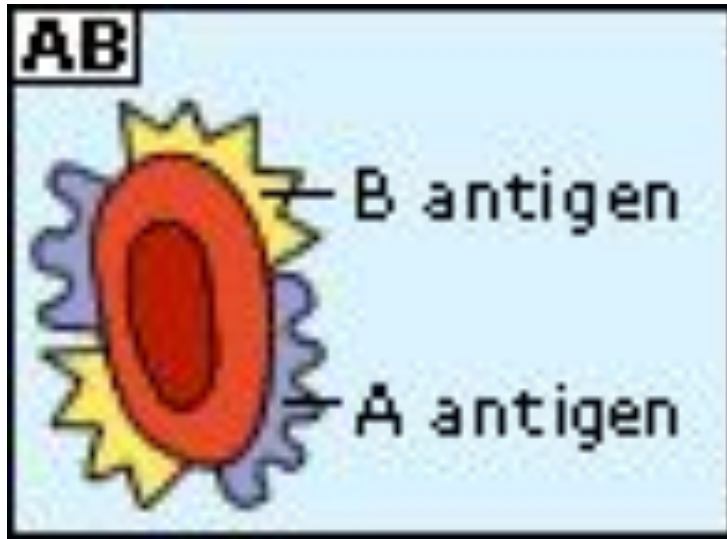
Blood group A

If you belong to the blood group A, you have A antigens on the surface of your RBCs and B antibodies in your blood plasma.



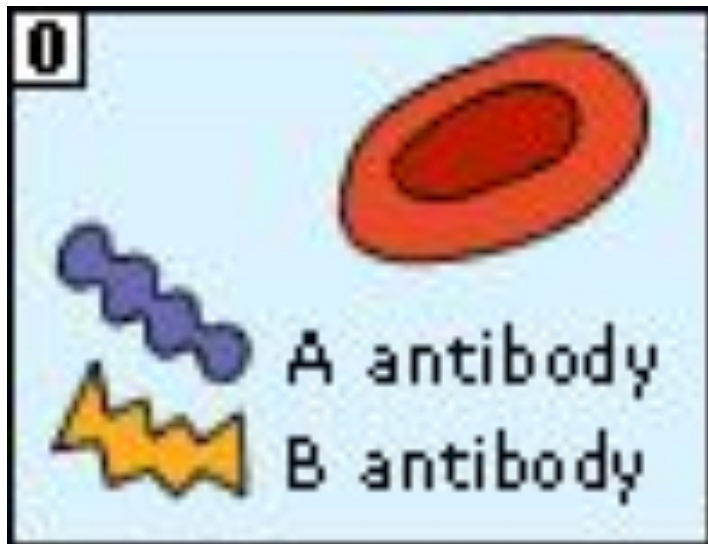
Blood group B

If you belong to the blood group B, you have B antigens on the surface of your RBCs and A antibodies in your blood plasma.



Blood group AB

If you belong to the blood group AB, you have both A and B antigens on the surface of your RBCs and no A or B antibodies at all in your blood plasma.



Blood group O

If you belong to the blood group O (null), you have neither A or B antigens on the surface of your RBCs but you have both A and B antibodies in your blood plasma.

Blood Type Compatibility

Blood type	Can Donate to	Can Receive from
A+	A+, AB+	A+, A-, O+, O-
A-	A+, A-, AB+, AB-	A-, O-
B+	B+, B-, AB+, AB-	B+, B-, O+, O-
B-	B+, B-, AB+, AB-	B-, O-
AB+(Universal Recipient)	AB+	All the groups
AB-	AB-	A-, B-, AB-
O+	A+, B+, AB+, O+	O+, O-
O- (Universal Donor)	All the groups	O-

Blood Group	Percentage
O ⁺	47
A ⁺	26
B ⁺	9
O ⁻	8
A ⁻	5
B ⁻	2
AB ⁺	2
AB ⁻	1

REQUIREMENTS

- Three glass slides
- Anti sera A
- Antisera B
- Antisera D with separate droppers and Lancet.

PRINCIPLE

The principle of the experiment is to find out the blood group by mixing the blood with the Antiserum and observe the agglutination by antigen antibody reaction.

OBSERVATION CHART

Antisera	Antisera A & 1 drop of blood	Antisera B & 1 drop of blood	Antisera D & 1 drop of blood
Agglutination			

Agglutination = +ve

No agglutination = -ve

RESULT:

- My blood group is

PRECAUTIONS:

- 1- Slides should be washed and dried carefully
- 2- Separate dropper should be used for each antisera.
- 3- The blood drop should not be touched to antisera.
- 4- A separate mixer should be used for mixing

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Relevant QUESTIONS?

Q-1-What are various blood groups?

There are four main blood groups in the ABO system: A, B, AB, and O. Each group can also be either RhD positive (+) or RhD negative (-), resulting in eight common blood types: A+, A-, B+, B-, AB+, AB-, O+, and O-.

Q-2- What is the basis of ABO blood group?

The ABO blood group system is based on the presence or absence of antigens A and B on the surface of red blood cells.

Q-3-Where is the agglutinin present?

Agglutinins are antigens present on the surface of red blood cells (RBCs). They are the substances that determine a person's blood type and can cause blood cells to clump together (agglutinate) when they react with specific antibodies in the blood.

Q-4-Where is the agglutinins present?

Agglutinins, which are antibodies that cause agglutination (clumping) of cells, are primarily found in the blood plasma or serum.

Q-5-How does the blood group help in finding the disputed parentage?

Blood groups can be used to exclude potential fathers in cases of disputed parentage, but they cannot definitively prove paternity. If a child has a blood type antigen that neither parent possesses, it definitively proves the man is not the father. However, if the blood types are compatible, it doesn't necessarily mean the man is the father, as other individuals could also share the same blood type.

Q-6-Who is universal donor and why ?

O blood group is the universal donor because the blood has no antigens present in it

Q-7-Who is universal recipient and why ?

Blood donors with type AB+ (AB positive) are referred to as universal recipients and can receive red blood cells from any other blood type.

Q-8-What are the indications of blood transfusion

- Indications of blood transfusion can be summed up as:
- Anemia.
- Major Surgical Operation.
- Accidents resulting in considerable blood loss.
- Cancer patients requiring therapy.
- Women in childbirth and newborn babies in certain cases.
- Patients of hereditary disorders like Haemophilia and Thalassaemia.
- Severe burn victims.

Q-9-What are the hazards of mismatched ABO blood transfusion

Mismatched ABO blood transfusions can lead to serious and potentially fatal complications. The primary hazard is a hemolytic transfusion reaction, where the recipient's immune system attacks the donor red blood cells, causing them to be destroyed. This can result in a range of adverse effects, including acute kidney injury, DIC, shock and even death.

Q-10-What are the expected ABO blood groups in offspring of the parents belonging to blood group O and A?

In the ABO blood group system, if one parent has blood type O and the other has blood type A, their children can have either blood type O or blood type A. This is because the gene for blood type O is recessive, while the gene for blood type A is dominant.

Q-11-What is the incidence of prevalence of Rh blood group?

The Rh blood group system is characterized by the presence or absence of the RhD antigen. Most people (around 85%) are Rh-positive, meaning they have the RhD antigen on their red blood cells. The remaining 15% are Rh-negative, lacking the antigen.

Q-12-What is the historical back ground for the Rh Blood Group System.

The Rh (rhesus) system was first recognized in 1939 and defined as D in 1940. The antibody produced by guinea pigs and rabbits in response to rhesus monkey cells agglutinated 85% of human RBCs. This was called anti-Rhesus, and the corresponding human antigen was called Rhesus.

Q-13-What are the main differences between ABO and Rh blood groups?

ABO System: It classifies blood on the basis of the presence or absence of antigen A and antigen B. Rh System: It classifies blood on the basis of the presence or absence of RhD antigen.

Q-14-Why is the second or successive mismatched Rh blood group transfusion more hazardous than the first?

A second or subsequent mismatched Rh blood transfusion is more hazardous than the first because the recipient's body has already produced antibodies against the Rh antigen during the initial exposure

Q-15-What is the expected RH blood group in the offspring when the parents belong to Rh positive and Rh negative blood groups

When parents have Rh positive and Rh negative blood types, the offspring's Rh blood group will be Rh positive in about half of the cases. If the father is Rh positive (either homozygous or heterozygous) and the mother is Rh negative, there is a 50% chance the child will be Rh positive and a 50% chance the child will be Rh negative

Q-16-Under what conditions erythroblastosis fetalis occur?

Erythroblastosis fetalis, also known as hemolytic disease of the fetus and newborn (HDFN), occurs when there is a blood group incompatibility between a mother and her fetus, leading to the destruction of fetal red blood cells. This typically happens when an Rh-negative mother carries an Rh-positive fetus.